Assignment05

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Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```r
# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic_Sampling_Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic_Sampling_Data-2026-03-10.xlsx"),
                   sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data",
 ↪  "Aquatic_Sampling_Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

Class code

```r
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

```r
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

```r
# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(med_ph = median(p_h, na.rm = TRUE),
```

```
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")
```

```
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
  # group by date, site, taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter (sum abundance divided by 7.5
   ↪  liters)
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
  # ungroup data frame
  ungroup() |>
  # create a new column called `date_site` to join with water quality
  unite("date_site", date, site, remove = FALSE) |>
  # join with water quality
```

```r
  left_join(water_quality_clean, by = c("date_site")) |>
  # join with taxon list
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # add full names of sites
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>
  # setting factor levels for site
  mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
   ↪  = TRUE),
         site_full = fct_rev(site_full))
```

```r
# base layer
salinity <- ggplot(data = aq_ins_clean,
       # x-axis: site
       # y-axis: mean salinity
       mapping = aes(x = site_full,
                     y = med_sal)) +
  # first layer: salinity measurements at each survey
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # second layer: points to represent means
  stat_summary(fun = median,
               color = "red",
               size = 1) +
  # relabelling x- and y-axes
  labs(x = "Site",
       y = "Median salinity (ppt)",
       title = "Site salinity") +
  # different theme
  theme_bw()
```

```
# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>
  # filter to only include copepods
  filter(taxon == "copepod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))
```

```
# base layer
copepods <- ggplot(data = copepods,
                   # x-axis: site
                   # y-axis: mean abundance/L
                   mapping = aes(x = site_full,
                                 y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
               fun = "median",
               size = 4,
               color = "blue") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Copepods") +
  # different theme
  theme_bw()
```

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

- x-axis: different sites
- y-axis: log + 1 transformed average ostracod abundance per liter
- geometry to represent average abundance/L: `geom_jitter()`, circle points
- geometry to represent medians: `stat_summary()`, circle points with a distinct color

b. Wrangle your data

```
# create new object from cleaned aquatic invertebrate data
ostracods_df <- aq_ins_clean |>
  # keep only ostracod rows
  filter(taxon == "ostracod") |>
  # log + 1 transform average abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

# display 10 random rows
ostracods_df |>
  slice_sample(n = 10)
```

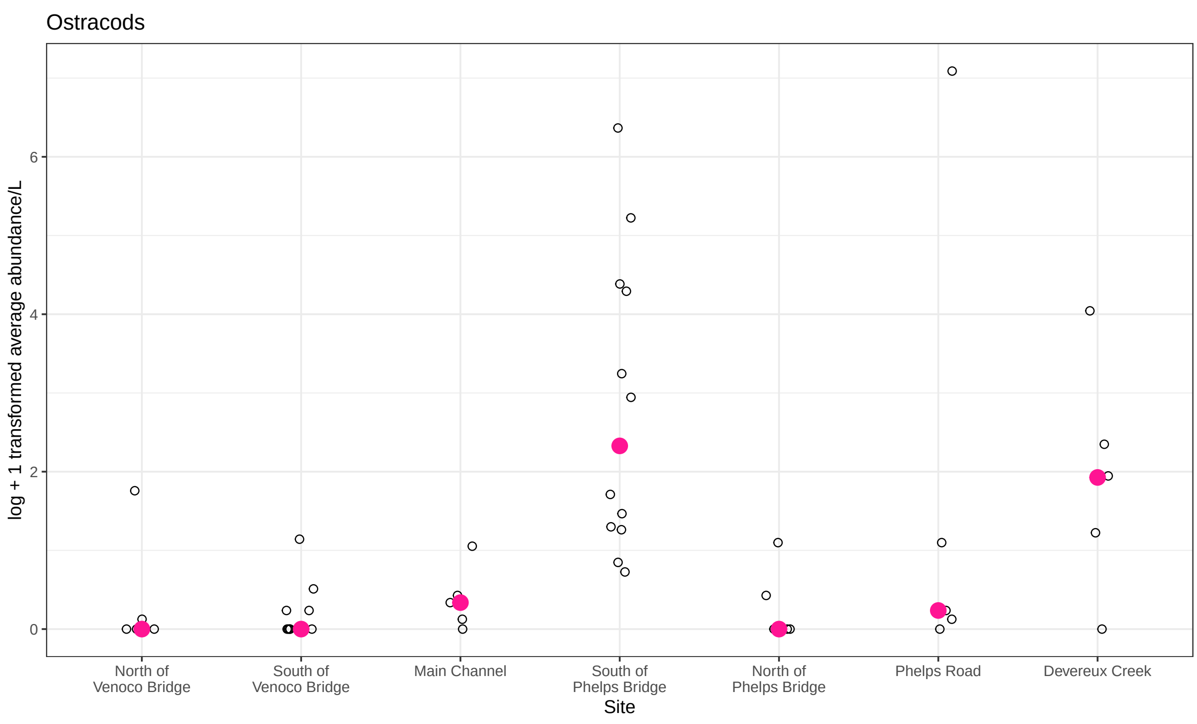
```
# A tibble: 10 x 24
   date_site       date                site  taxon ave_lit med_ph med_sal med_do
   <chr>           <dttm>              <chr> <chr>   <dbl>  <dbl>   <dbl>  <dbl>
 1 2021-09-01_NPB2 2021-09-01 00:00:00 NPB2  ostr~    0     NA      NA     NA
 2 2022-02-23_NVBR 2022-02-23 00:00:00 NVBR  ostr~    0      8.86   45.2   10.2
 3 2022-11-12_NEC  2022-11-12 00:00:00 NEC   ostr~    0      7.33    7.01   2.4
 4 2022-05-06_NMC  2022-05-06 00:00:00 NMC   ostr~    0      9.96    1.45  16.5
 5 2022-11-19_NPB1 2022-11-19 00:00:00 NPB1  ostr~    2      7.26    2.47   3.96
 6 2021-11-16_NPB  2021-11-16 00:00:00 NPB   ostr~   72.1    7.3     2.38  12.1
 7 2021-11-12_NPB  2021-11-12 00:00:00 NPB   ostr~   79.2   NA      NA     NA
 8 2022-02-21_NPB  2022-02-21 00:00:00 NPB   ostr~    1.33   7.39   34.1    0.34
 9 2023-02-28_NPB1 2023-02-28 00:00:00 NPB1  ostr~    0     NA      NA     NA
10 2022-02-25_NEC  2022-02-25 00:00:00 NEC   ostr~    0      8.75   44.1    9.87
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code your figure

```
# base layer
ostracods <- ggplot(data = ostracods_df,
                    # x = site, y = log abundance/L
                    mapping = aes(x = site_full,
                                  y = log_ave_lit)) +
  # first layer: individual survey points
```

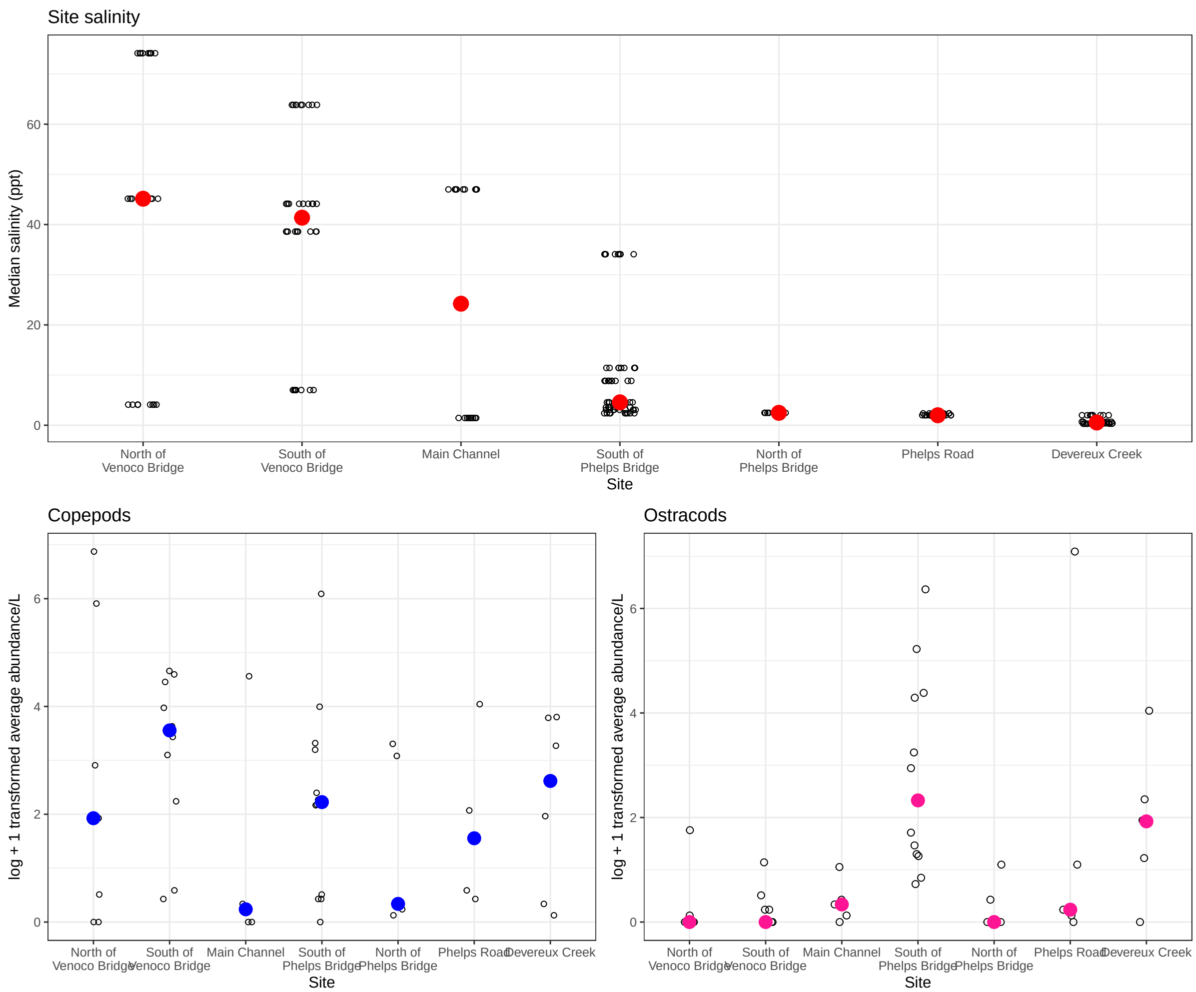
```
  geom_jitter(height = 0,
              width  = 0.1,
              shape  = 21,
              size   = 2) +
  # second layer: median per site, distinct in size and color
  stat_summary(fun   = "median",
               geom  = "point",
               color = "deeppink",
               size = 4) +
  # wrap long site labels for readability
  scale_x_discrete(labels = label_wrap(15)) +
  # readable axis labels and taxon-specific title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Ostracods") +
  # non-default theme
  theme_bw()

# display object
ostracods
```



d. Create a multipanel figure

```
# salinity on top, copepods and ostracods side-by-side below
(salinity) / (copepods | ostracods)
```



e. Interpret your figure

- Differences between sites in copepod and ostracod abundance: Copepod medians are the blue circles on the bottom left panel. They are generally non-zero at every site. They have largest at South of Venoco Bridge (about 3.5) and the lowest at Main Channel and North of Phelps Bridge (~0.3). Ostracod medians are the pink circles on the bottom right panel. They are essentially zero at the two Venoco Bridge sites and at North of Phelps Bridge. They have sharply rised at South of Phelps Bridge and Devereux Creek.

- How ostracod abundance may relate (or not) to site-level differences in salinity: Because sites are ordered from highest to lowest median salinity left-to-right in the top panel, the pink ostracod medians generally rise as the red salinity medians fall. Ostracods are zero where median salinity exceeds 40 ppt and reach their highest values at sites with median salinity below 5 ppt. The relationship isn't very corelated. Main Channel, North of Phelps Bridge, and Phelps Road all have low ostracod medians despite moderate or low salinity, so salinity alone doesn't fully explain the site-level pattern.